

CURRICULUM VITAE

David T. Blaauw

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I Personal Data

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II Employment History

A. Education

Doctor of Philosophy in Computer Science, University of Illinois, Urbana-Champaign, January 1992.
Thesis: “Functional Abstraction in Switch-Level Simulation.”
Advisor: Professor Jacob A. Abraham

Master of Science in Computer Science, University of Illinois, Urbana-Champaign, May 1989.
Thesis: “Automatic Generation of Behavioral Models.”
Advisor: Professor Jacob A. Abraham

Bachelor of Science in Physics with a second major in Computer Science, Duke University, May 1986.

B. Present Position

Professor of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, Michigan.

C. Employment History

- September 2007 - Present: Professor, Department of Electrical Engineer and Computer Science, University of Michigan, Ann Arbor, Michigan
- August 2001 - September 2007: Associate Professor, Department of Electrical Engineer and Computer Science, University of Michigan, Ann Arbor, Michigan.
- September 1994 - August 2001: Engineering Manager, Advanced Design Technology, Motorola, Inc., Austin, Texas.
- August 1993 - September 1994: Staff engineer, Semiconductor Systems Design Technology Group, Motorola, Inc., Austin, Texas.
- August 1992 - August 1993: Development Staff Member, IBM Corporation, Endicott, New York.

D. Honors and Awards

- Paper selected by the Communication of the ACM (CACM) as a Research Highlights (RH), Yufeng Gu, Arun Bala Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath N Paul, Md Vasimuddin, Sanchit Misra, David Theodore Blaauw, Satish Narayanasamy, Reetuparna Das, “GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis,” Communication of the ACM (CACM), October 2024
- Best Paper Award, Pierre Abillama, Zichen Fan, Yu Chen, Hyochan An, Qirui Zhang, Seungkyu Choi, David Blaauw, Dennis Sylvester, Hun-Seok Kim, “SONA: An Accelerator for Transform-Domain Neural Networks with Sparse-Orthogonal Weights,” 34th IEEE International Conference on Application-specific Systems, Architectures, and Processors (ASAP), July 2023
- Selected for the ISCA 25 years retrospective (1996-2020) for year 2018: “Neural cache: bit-serial in-cache acceleration of deep neural networks,” June 2023
- IEEE International Solid-State Circuits Conference Author-Recognition Award for its first 70 years as a contributor with more than 60 Papers.
- ACM/IEEE International Symposium on Microarchitecture (MICRO) Hall of Fame for co-authoring 8+ papers in Micro, October 2022
- IEEE International Symposium on Computer Architecture (ISCA) Hall of Fame for co-authoring 8+ papers in ISCA, 2022
- Best Paper Award, Andrea Bejarano-Carbo, Hyochan An, Kyojin Choo, Shiyu Liu, Dennis Sylvester, David Blaauw, Hun Seok Kim, “Millimeter-Scale Ultra-Low-Power Imaging System for Intelligent Edge Monitoring,” TinyML conference, March 2022
- Highlighted paper in Nature Electronics: “Actuating sub-millimetre robots” in Proc. 2022 IEEE Int. Solid-State Circuits Conference, March 2022
- MobiCom 2021 Best Paper award “mSAIL: milligram-scale multi-modal sensor platform for monarch butterfly migration tracking”. Also selected to be highlighted for the combined edition of Fall 2021/Spring 2022 SIGMOBILE Research Highlights
- 2022 Distinguished Academic Achievement Alumni Award from the Department of Computer Science at the University of Illinois at Urbana-Champaign. One award given per year.
- 2022 IEEE Micro Top Picks Special Issue on Top Picks from the 2021 Computer Architecture Conferences, Honorable Mention, “SquiggleFilter: An Accelerator for Portable Virus Detection”
- Test of Time award, “the highest honor an academic paper receives for its impact and recognizes an influential MICRO paper whose influence is still felt 18-22 years after its initial publication” for

- “Razor: A low-power pipeline based on circuit-level timing speculation,” IEEE/ACM International Symposium on Microarchitecture, 2003, October, 2021
- Winner, 2019 Monarch Butterfly Fund for creating a system to track the flight of individual monarch butterflies on their migration to Mexico
 - 2019 Distinguished University Innovator Award given to one team in the university for developing and marketing transformative ideas and technologies
 - Named the Kensall D. Wise Collegiate Professor of Electrical Engineering and Computer Science, 2019
 - Best Student Paper Award, “A 4×4×4-mm³ Fully Integrated Sensor-to-Sensor Radio using Carrier Frequency Interlocking IF Receiver with -94 dBm Sensitivity,” IEEE Radio Frequency Integrated Circuits Symposium (RFIC), June 2019
 - 2019 IEEE Micro Top Picks special issue on the Computer Architecture Conferences, “Neural Cache: Bit-Serial In-Cache Acceleration of Deep Neural Networks”
 - 15 year retrospective most influential paper in ISCA 2002 award for groundbreaking research in power-efficient computing, ACM/IEEE International Conference on Computer Architecture (ISCA), 2017
 - Member of University of Illinois Engineering Advisory Panel. 2015 through current.
 - Ranked as the top publishing author at IEEE VLSI Circuits Symposium over the last 30 years of the conference with 38 publications. June 2017
 - 2016 University Researcher Award, Semiconductor Industry Association (SIA) – Semiconductor Research Corporation (SRC), established by the semiconductor industry association to recognize lifetime research contributions to the U.S. semiconductor industry by university faculty. One award given per year for circuits and technology each.
 - 2016 IEEE Micro Top Picks special issue "MBus: The Missing Interconnect that Enables the Modular Millimeter-Scale Computing Class and Connects the World's Smallest Computer,"
 - Best Paper Award, “Racetrack Converter: A Low Power and Compact Data Converter Using Race-track Spintronic Devices,” IEEE International Symposium on Circuits and Systems (ISCAS), May 2015
 - 2014 John von Neumann Student Research Award for Excellence in Systems Research – SONIC Annual Review Meeting
 - Recognized as top 50 innovator over the last 50 years graduating from the University of Illinois EECS department in 2014
 - College of Engineering Innovation Excellence Award for 2013-2014
 - Design Automation Conference (DAC) 50th Anniversary award for being the top 10 most cited DAC authors in DAC’s 50 year history, June 2013
 - Design Automation Conference (DAC) 50th Anniversary award for publishing the most papers in the fifth decade of DAC’s history, June 2013
 - 2013 University of Michigan Electrical Engineering and Computer Science (EECS) Department Outstanding Achievement Award for innovative research in variation-tolerant and energy efficient integrated circuit design, and exceptional mentoring and teaching in the area of VLSI circuits
 - International Solid-State Circuits Conference (ISSCC) 60th Anniversary Special Recognition top 10 contributing author over the last 10 years, February 2013

- IEEE/ACM International Conference on Computer-Aided Design (ICCAD) Ten Year Retrospective Most Influential Paper Award, “Combined Dynamic Voltage Scaling and Adaptive Body biasing for Lower Power Microprocessors under Dynamic Workloads,” ICCAD 2002 Conference, November 2012
- Second Prize in the 18th Samsung Human-Tech Thesis Competition for research on millimeter sensor design, February 2012
- IEEE Fellow status, January 2012
- Winner MuSyC Research Consortium annual best poster award, “A Modular 1mm³ Die-Stacked Sensing Platform,” Nov 2011
- Winner 11th International VLSI-Symposium Low Power Design Contest, “SWIFT: A 2.1 Tb/s 32x32 Self-Arbitrating Manycore Interconnect Fabric,” IEEE Symposium on VLSI Circuits (VLSI-Symp), June 2011
- Winner Design Automation Conference (DAC)/International Solid-State Circuits Conference (ISSCC) Design contest, “Design and Implementation of Centip3De, a 7-layer Many-Core System,” Design Automation Conference (DAC)/International Solid-State Circuits Conference (ISSCC), Feb/June 2011
- Best Paper Award, “Low Power Circuit Design Based on Heterojunction Tunneling Transistors (HETTs),” ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED), August 2009
- 2008 Ted Kennedy Family Team Excellence Award (award shared with Todd Austin, Scott Mahlke, Trevor Mudge, Marios Papaefthymiou). The Ted Kennedy Family Team Excellence Award is an annual award given by the University of Michigan, College of Engineering that recognizes the production of an extraordinary and significant piece of work from current or recent collaboration in teaching or research to the College of Engineering.
- 2008 Richard Newton GSRC Industrial Impact Award for “development of the Razor technology” (award shared with Professor Todd Austin). The Richard Newton GSRC Industrial Impact Award is an annual award given by the GSRC DARPA/MARCO center that recognizes research that is “at least five years old and has had a significant industrial impact.”
- University of Michigan College of Engineering Research Excellence Award for 2007-2008, January 2008
- Best Paper Nomination, “Energy Efficient Near-threshold Chip Multi-processing,” ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED), August 2007
- Best Paper Nomination, “Self-timed Regenerators for High-speed and Low-power Interconnect,” ACM/IEEE International Symposium on Quality Electronic Design (ISQED), March 2007
- Microprocessor Review Analysts’ Choice Award in Innovation for “Introducing Speculation on Correctness as a Method for Allowing Circuit Operation Beyond Worst-Case Design,” Microprocessor Review, February 2007
- 2004 IEEE Micro Top Picks special issue on the most industry relevant and significant papers of the year in computer architecture, “Razor: Circuit-Level Correction of Timing Errors for Low-Power Operation”
- University of Michigan Henry Russel Award for “Exceptional Scholarship and Conspicuous Ability as a Teacher,” November 2004
- Best Paper Nomination, “Parametric Yield Estimation Considering Leakage Variability,” ACM/IEEE Design Automation Conference (DAC), June 2004

- Best Paper Award, “Razor: A Low-Power Pipeline Based on Circuit-Level Timing Speculation,” ACM/IEEE International Symposium on Microarchitecture (MICRO), November 2003
- Best Regular Paper Award, “Noise Analysis Methodology for Partially Depleted SOI Circuits,” IEEE Custom Integrated Circuits Conference (CICC), September 2003
- IBM Faculty Award, IBM Center for Advanced Studies, June 2003
- Best Paper Award, “Statistical Delay Computation Considering Spatial Correlations,” ACM/IEEE Asia-Pacific Design Automation Conference (ASP-DAC), January 2003
- IBM Faculty Award, IBM Center for Advanced Studies, June 2002
- Best Paper Nomination, “Pre-route Noise Estimation in Deep Submicron Integrated Circuits,” ACM/IEEE International Symposium on Quality Electronic Design (ISQED), March 2002
- Best Paper Nomination, “Driver Modeling and Alignment for Worst-Case Delay Noise,” ACM/IEEE Design Automation Conference (DAC), June 2001
- Best Paper Award, “On-Chip Inductance Modeling and Analysis,” ACM/IEEE Design Automation Conference (DAC), June 2000
- Motorola Innovation Award, 1997
- Motorola High Impact Technology Award, 1996

III Research Experience

A. Research Interests

My research interests focus on high-performance and low-power VLSI circuits, particularly addressing nano-meter design issues pertaining to power, performance and robustness. My aim is to develop novel circuit design techniques for effective VLSI design in the nano-meter era, in conjunction with efficient and accurate analysis and optimization methods for large, multi-million transistor designs.

B. Doctoral Students Supervised

<u>Student</u>	<u>Thesis Title/Topic</u>	<u>Graduation Date</u>
Olivia Lee		In Progress
Rahul Narasimha		In Progress
Hyunwon Chung		In Progress
Mohammed Ashfakh		In Progress
Jeongtaek Chang		In Progress
Joey Costello		In Progress
Yi Shen		In Progress
Jungho Lee	Neural Stimulation In progress	In Progress
Andrea Bejarano	mm-Scale Image Sensing Through Low Power Circuit and Algorithm Co-Design	In Progress
Yichen Gu	Computational Genomics for RNA Velocity Modeling	In Progress
Chien-Wei Tseng	Ultra-Low Power RF-Localization for Asset Management	March 2024
Kuan Yu Chen	Design and Implementation of Domain-Specific Programmable Spatial Accelerators	November 2023

Rohit Rothe	Energy Efficient Circuit Design Techniques and Beyond CMOS Exploration for Internet of Things (IoT)	November 2023
Zhen Feng	Energy-Efficient Low-Power Mm-Scale Wireless Communication System	November 2022
Jihwan Seol	Energy-Efficient Mixed-Signal Circuits and Systems for Communication and Signal Processing	November 2022
Zhehong Wang	Application of New Technologies to Neural Networks Processors	December 2021
Xiao Wu	Energy Efficient Circuits and System for Internet of Things and Hardware Accelerator Design for Genome Sequencing	July 2019
Li-Xuan Chuo	Miniaturized Low-Power and Energy-Efficient RF Wireless Communication and Sensing Systems	June 2019
Ziyun Li	Energy-Efficient Mobile Computer Vision and Machine Learning Processors	May 2019
Kyojin Choo	Charge-domain analog/mixed-signal circuits and applications	Sept 2018
Yao Shi	Millimeter-Scale and Energy-Efficient RF Wireless System	Aug 2018
Wootack Lim	Ultra-low Power Circuit Design for Miniaturized IoT Platform	May 2018
Taekwang Jang	Circuit and System Designs for Millimeter Scale IoT and Wireless Neural Recording	Dec 2017
Wanyeong Jung	Low-Power Energy Efficient Circuit Techniques for Small IoT Systems	April 2017
Supreet Jeloka	Cross-point Circuits for Computation, Interconnects, Security and Storage	Jan 2017
Yejoong Kim	Robust Circuit Design for Low-Voltage VLSI	May 2015
Nathaniel Pinckney	Near-Threshold design	July 2015
Dongmin Yoon	Low power timer references	Jan 2015

Inhee Lee	Power management for ultra-low power sensors systems	Oct 2014
Gyouho Kim	Ultra-low power visual monitoring	Aug 2014
Bharan Giridhar	Adaptive Computing	Dec 2013
Zhi Yoong Foo	Low power processor design techniques	Aug 2013
Sudhir Satpathy	Fast and low power interconnect fabrics	Dec 2012
David Fick	Adaptive Low-power design	Aug 2012
Yoonmyung Lee	Ultra Low-Power Memory Design	April 2012
Prashant Singh	Reliability analysis and wear-out detection	April 2010
Nurrachman Liu	Automatic tuning of VLSI circuits	April 2010
Brian Cline	Process variation modeling for advance semiconductor circuits	Feb 2010
Cheng Zhuo	VLSI wearout modeling	Dec 2010
Ravikishore Gandikota	Crosstalk-Noise analysis for nanometer VLSI circuits	Aug 2009
Carlos Tokunaga	Circuits and architectures for secure processing	Sep 2008
Shidhartha Das	Razor: circuit speculation for power and performance efficient design	Oct 2008
Kaviraj Chopra	Statistical timing analysis including spatial correlations	Apr 2008
Eric Karl	Reliable computing on unpredictable silicon	Mar 2008
Sanjay Pant	Power grid analysis and design	Dec 2007
Mini Nanua	Leakage and noise analysis in nano-scale technologies	Apr 2007

Bo Zhai	Dynamic voltage scaling for embedded processor designs	Mar 2007
Rajeev Rao	Modeling and design of low-power VLSI systems under for multiple sources of uncertainty	Jul 2006
Dongwoo Lee	Analysis and minimization of leakage current	May 2005
Aseem Agarwal	Statistical timing analysis for VLSI circuits	Mar 2005

C. Masters Students Supervised

<u>Student</u>	<u>Thesis Title/Topic</u>	<u>Graduation Date</u>
Alhad Daftardar	High Performance Computing	In Progress
Peijun Hou	Capacitance to Digital Conversion	In Progress
Li Yu Chen	High Voltage Upconversion	May 2021
Hengfei Zhong	Power Circuit Design	May 2020
Ashwin Bhat	Error Tolerant Long-Read Alignment for Genomic Sequencing	April 2020
Hyungjoo Seo	Low Power Energy Harvesting	May 2019
Tim Wesley	Low Power Neural Network Accelerators	May 2019
Yu Zeng	Low Power Crystal Oscillator Circuits	Dec 2017
Dongkwun Kim	Low Power Voltage References	May 2017
Skyler Skrzyniarz	Low Power Correlation Architectures for GPS	June 2015

Junhua Gu	Low Power Circuits for Analog to digital interfaces	May 2015
Ruochen Xie	Energy Reduction of FeRAM Memories for Millimeter Sensors	May 2015
Allen Wang	Low Power Level Conversion	May 2014
Naveen Akesh	Low Power Audio Device for Developing World	May 2014
Zhe Yu	RF Communication for Millimeter Scale Sensors	May 2014
Hsi-Shou Wu	Low Power Word-Spotting	April 2014
Siddharth Saxena	Low power correlation circuits	May 2013
Karan Jain	Low power synchronization using ambient RF signals	April 2012
Jordan LeNoach	pH sensor for millimeter sensors	Dec 2011
Jeffrey Yeh	Chip design for the developing world	April 2011
Nate Robert	Low power LDO	Dec 2010
Vikas Vinay	Low power Class-D amplifier for developing world applications	Dec 2010
Abhishek Madhavan	Low power chip design	Dec 2010
Junsun Park	Intra-cellular chip design	May 2010
Jou-ching (George) Sung	Low power ADC design	Aug 2009
Mao-Ter Chen	Low power sensor node design	Dec 2008
Sudharsan Kalaiselvan	Razor-3: A circuit speculation and SEU tolerant circuit technique	May 2007
Deepesh John	Low power design through typical-case optimization	May 2006

Yueh-Chuan Tzeng	Encryption processor for side channel attack avoidance	May 2006
Meghna Singhal	Low power design using subthreshold operation	May 2006
Amir Borna	Analysis of lithographic variations for chip performance	Aug 2005
Amit Jain	Delay modeling for non-ramp input transitions	Nov 2004
Toan Pham	Clock skew reduction using Razor flip-flops	Dec 2003
Bhavana Thudi	Non-iterative switching window computation for delay noise	May 2003
Wesley Kwong	Efficient circuit-level analysis of gate-oxide tunneling current in VLSI designs	May 2003

D. Research Grants

- WoodNext Foundation, “Self-Powered Sensor using Graphene Energy Harvesting,” \$210,000, PI: Blaauw, 1/1/24-1/8/2029
- Semiconductor Research Corporation (SRC), Inc., “CogniSense: Center on Cognitive Multispectral Sensors,” \$4,000,000, with \$1,300,000 to Blaauw, 1/1/2023-12/31/2027
- DARPA “DR DASH,” \$3,072,000, with \$1,100,000 to Blaauw, PI: Hun Seok Kim, 6/1/2023-11/1/2026
- National Institutes of Health (NIH), “Optical neural motes to enable high density recording through intact dura in a nonhuman primate,” \$2,572,751, with \$700,000 to Blaauw, PI: David Blaauw, 8/18/2022-7/31/2025
- DARPA, Domain-focused Advanced Software-Reconfigurable Heterogeneous System-on-Chip (DASH-SoC), \$5,152,477 with \$1,300,000 to Blaauw, PI: David Blaauw, 7/18/2018-5/31/2025
- National Science Foundation (NSF), “Collaborative Research: IIBR Multidisciplinary: mSAIL (Michigan Small Animal Integrated Logger): milligram-scale, multi-modal sensor and analytics monitoring platform,” \$1,200,000 with \$613,589 to Blaauw, PI: Delbert Andre Green, 4/15/2021-3/31/2025
- DARPA, “Configurable Optical Communications via Heterogeneous-Processing Optimized Node (COCHON),” \$1,100,000 with \$500,000 to Blaauw, PI: David Blaauw, 12/14/2021 - 3/28/2025
- Delaware Nation Industries (DNI), “Smart ePants,” \$974,000 to Blaauw, PI: David Blaauw, 7/26/2023 - 12/31/2024
- Ministry of Defense “M3 Audio-Logger,” \$1,561,110, with \$1,000,000 to Blaauw, PI: David Blaauw, 2/18/2019 - 8/31/2024

- Ministry of Defense “Low Earth Orbit Based Localization,” \$1,326,467, with \$890,000 to Blaauw, PI: David Blaauw, 3/12/2020 - 6/30/2024
- City Labs, Inc., “High-Voltage Down Conversion in Support of High-Voltage Batteries,” \$280,000, PI: David Blaauw, 6/30/2021 - 6/29/2024
- University de Franche-Comte, “Assembly, Testing and Delivery of High-Voltage Micro-Motes for Programmable Matter,” \$259,033, PI: David Blaauw, \$12/1/2022 - 6/8/2024
- Advanced Energy Consortium (AEC), “Millimeter Scale Down-hole Sensors,” \$650,000, with \$400,000 to Blaauw, PI: David Blaauw, 3/1/21 – 5/31/24
- University of Pennsylvania / National Science Foundation (NSF), “NRI: General Purpose Microscopic Robots via Integrated Electronics,” \$1,200,000 with \$300,000, to Blaauw, PI: Marc Miskin, 9/1/2022 - 8/31/2023
- National Geographic Society, “M3 Monarch Migration Study,” \$150,000 with \$75,000 to Blaauw, PI: David Blaauw, 10/11/2019-10/31/2023
- National Science Foundation (NSF), “StiMote: An Ultrasmall, Modular Neurophtonic Stimulator,” \$649,130, with \$216,376 to Blaauw, PI: James Weiland, 06/24/2021 – 6/23/2023
- Intel Corporation, “Valleytronics Circuit Design for Efficient Realization of Computational Logic,” \$555,000, PI: David Blaauw, 6/31/2020-6/30/23
- DARPA “Datalink Applications of DSSoC-DASH (KESTREL),” \$720,000 with \$360,000 to Blaauw, PI: Hun Seok Kim, 1/1/2020-1/17/2023
- Ministry of Defense, “Low Power Localization,” \$840,000, with \$420,000 to Blaauw, PI: David Blaauw, 3/12/2020-3/31/2022
- Semiconductor Research Corporation (SRC), “Analog and Digital Assist Techniques to Improve Mixed-Signal,” \$130,000 with \$65,000 to Blaauw, PI: Dennis Sylvester, 9/1/2020 - 12/31/2021
- Facebook, “Context-Aware Multi-Sensor Fusion System and SoC Integrated Circuits and Digital Signal Processing,” \$239,752, with \$119,876 to Blaauw, PI: Dennis Sylvester, 1/1/20-12/31/21
- ARM, “ULP Sensor Fusion SoC for Micro Robots and Tiny IoT,” \$320,000 with \$160,000 to Blaauw, PI: David Blaauw, 9/1/19-8/31/21
- Sony Corporation of America, “Ultra Low Power intelligent imaging sensors for the Internet of Things,” \$1,493,433, with \$511,041 to Blaauw, PI: Dennis Sylvester, 9/1/19-8/31/22
- Ministry of Defense “Ultra-Miniature Imager Technical Demonstrator,” \$1,057,762, with \$657,762 to Blaauw, PI: David Blaauw, 7/18/19-3/31/2021
- Ministry of Defense “M3 Audio-Logger,” \$1,204,352, with \$704,352 to Blaauw, PI: David Blaauw, 2/18/19-3/31/2021
- Semiconductor Research Corporation (SRC), “Mixed-Signal Circuits Enabling Ultra-Low Power Wireless Sensors,” \$270,000, with \$139,995 to Blaauw, PI: David Blaauw, 1/1/18-12/31/20
- Advanced Energy Consortium (AEC), “Millimeter Scale Down-hole Sensors,” \$650,000, with \$400,000 to Blaauw, PI: David Blaauw, 3/1/18-5/1/24
- National Science Foundation (NSF), “NCS FR - Elucidating the relationship between motor cortex neural firing rates and dextrous finger movement EMG for use in brain computer interfaces,” \$2,276,395, with \$194,603 to Blaauw, PI: Cindy Chestek, 9/1/19-8/31/23
- DARPA, “Domain-focused Advanced Software-reconfigurable Heterogeneous System-on-Chip (DASH-SoC),” \$2,420,233, with \$1,318,078 to Blaauw, PI: Hun-Seok Kim, 7/18/18-6/30/22

- National Science Foundation (NSF), “SHF: Medium: Compute Caches: Opportunistic Extreme Scale Parallelism in General Purpose Processors,” \$300,000, with \$80,579 to Blaauw, PI: Reetuparna Das, 10/1/18-9/30/20
- Leidos, “Trestle – small acoustic sensors” \$1,882,300, with \$983,104 to Blaauw, PI: David Blaauw, 5/3/19-2/28/2021
- National Institute of Health (NIH), “A 100 μ m Scale Single Unit Neural Recording Probe Using IR-Based Powering and Communication,” \$429,971, with \$224,890 to Blaauw, PI: David Blaauw, 9/1/18-8/31/2021
- DARPA, “Transmuter: A Reconfigurable Computer,” \$5,845,690, with \$1,524,096 to Blaauw, PI: Ron Dreslinski, 8/6/18-3/30/21
- DARPA, “OpenROAD: Foundations and Realization of Open, Accessible Design Integrated Circuits and Digital Signal Processing,” \$1,600,000, with \$248,754 to Blaauw, PI Andre Kahng, 6/1/18-5/31/22
- DARPA, “Fully-Autonomous SoC Synthesis using Customizable Cell-Based Synthesizable Analog Circuits,” \$3,199,021, with \$482,255 to Blaauw, PI: David Wentzloff, 6/12/18-6/11/20
- Leidos, “NZero Acoustic Switch (Dorado),” 930,000, with \$699,424 to Blaauw, PI: David Blaauw, 11/11/17-6/30/19
- National Institute of Standards and Technology (NIST), United States Department of Commerce (DoC), “Decimeter Accurate, Long Range Non-Line-of-Sight RF Localization Solution for Public Safety Applications,” \$997,873, with \$310,041 to Blaauw, PI: Hun Seok Kim, 6/1/17-5/31/21
- Ministry of Defense, “Highly Size Constrained Logging Sensor Development,” \$2,580,843, with \$1,131,882 to Blaauw, PI: David Blaauw, 10/01/16-08/15/18
- Sony Electronics, Inc., “Low Power Motion Detection for the Internet of Things,” \$1,193,000, with \$400,000 to Blaauw, PI: Dennis Sylvester, 8/1/2016-12/31/2019
- DARPA, “Support for Porting S2CFF Design and Near-Threshold Voltage (NTV) Design Methodology to 32nm CMOS,” \$310,829 to Blaauw, PI: David Blaauw, 12/8/2015-8/31/2017
- DARPA, “Near Zero-Power, Continuous Acoustic Sensing Microsystem using Active Integrated Circuits and Digital Signal Processing,” \$2,505,000, with \$835,000 to Blaauw, PI: Dennis Sylvester, 9/28/2015-7/31/2019
- Semiconductor Research Corporation (SRC), “Infrared-Based Power Delivery and Communication for Implanted Sensors,” \$375,000 to Blaauw, PI: David Blaauw, 6/1/2015-5/31/2018
- Advanced Energy Consortium (AEC), “An Autonomous Microsystem Test-Bed for Extreme Environments: Strategic Options and Scalability Limits,” \$1,170,000, with \$585,000 to Blaauw, PI: David Blaauw, 1/1/2015-7/1/2017
- ARM Ltd, “ARM III: Research into Low Energy Computer Systems,” \$3,000,148, with \$555,555 to Blaauw, PI: Trevor Mudge, 2015-2018
- National Institute of Health (NIH), “SCH: INT: Wireless Implantable Electronic Biosensors for Tumor Monitoring,” \$1,759,460 with \$849,804 to Blaauw, PI: David Blaauw, 9/22/2014 - 8/31/2018
- Ministry of Defense, “M3 mm scale computing GPS logger,” \$1,352,145, with \$250,000 to Blaauw, PI: David Blaauw, 04/01/14 - 03/31/16
- Ministry of Defense, “Dstl Sensor Development Kits,” \$400,000, with \$112,500 to Blaauw, PI: David Blaauw, 12/13/13 - 3/17/14
- Ministry of Defense, “Architectural Design Study for M3 mm Scale Computing GPS Logger,” \$400,000, with \$175,000 to Blaauw, PI: David Blaauw, 8/15/2013-3/14/2014

- National Science Foundation (NSF), “SHF: Small: Minimally Invasive Error Detection/Correction for Runtime Margin Elimination,” \$450,000, with \$252,750 to Blaauw, PI: David Blaauw, 7/2012-6/2015
- BAE Systems/United States Army, “Center for Objective Microelectronics and Biomimetic Adaptive Technology (COM-BAT),” \$400,000, with \$135,000 to Blaauw, PI: Kamal Sarabandi, 9/2013-8/2016
- DARPA, “The TerraSwarm Research Center,” \$4,234,183, with \$1,624,897 to Blaauw, PI: Edward Lee, 1/15/2013-12/31/2017
- ARM, Ltd, “Low Power Computing for Embedded Applications,” \$5,000,000, with \$925,000 to Blaauw, PI: Trevor Mudge, 5/2010-5/2015
- Advanced Energy Consortium (AEC), “An Autonomous Microsystem Test-Bed for Extreme Environments: Integrating Sensor Elements, Electronics, and Packaging,” \$950,000, with \$226,625 to Blaauw, PI: Yogesh Gianchandani, 6/2012-12/2014
- Semiconductor Research Corporation (SRC), “Fast Power Supply Boosting for Energy-Efficient, High-Performance Processors,” \$360,000, with \$180,000 to Blaauw, PI: David Blaauw 8/2012 - 7/2015
- DARPA, “Systems on Nanoscale Information Fabrics (SONIC) Center,” \$7,008,335, with \$1,401,667 to Blaauw PI: Naresh Shanbhag, 1/2013-10/2017
- Ministry of Defense, “MM scale computing for GPS logger,” \$400,000, with \$175,000 to Blaauw, PI: David Blaauw
- Isocline Engineering LLC, “Power Efficient Software Define Radio (SDR) Mobile Architecture Technology for Handheld Devices,” \$220,093 to Blaauw, PI: David Blaauw, 03/01/2013 - 02/28/2015
- Isocline Engineering LLC, “Programmable Microchip for Accelerating Neuromorphic Object Recognition,” \$45,715 to Blaauw, PI: David Blaauw, 07/01/2013 - 12/31/2013
- QUALCOMM, “Near Threshold Computing,” \$100,000, gift, 8/2011-8/2013
- Oracle, “High Performance Razor Architecture” \$80,000, gift, 8/2013-8/2014
- AMD, “*In Situ* Wearout Detection and Mitigation,” \$100,000, gift, with \$50,000 to David Blaauw, 11/2011
- Food and Drug Administration, “Smart Rapid Palatal Expander for Pediatric Cleft and Palate Patients,” \$312,000, with \$136,000 to Blaauw, PI: Jeanne Nervina, 9/2011-8/2013
- National Science Foundation (NSF), “Integrating Circuits, Sensing, and Software to Realize the Cubic-mm Computing Class,” \$2,533,000, with \$519,265 to Blaauw, PI: David Wentzloff, 08/2011 - 7/2016
- Qualcomm, “Near-Threshold Computing,” \$50,000, gift, PI: David Blaauw, 05/2011
- Department of Energy, “Hardware-Software Co-Design for Non-Volatile Memory in Exascale Systems,” \$525,000 with \$202,747 to Blaauw, PI: Trevor Mudge, 1/2011-12/2013
- Intel Corporation, “A Confidence-Driven Model for Predictable Computing in Future Technologies,” \$249,000 with \$65,916 to Blaauw, PI: Zhengya Zhang, 1/2010-10/2010
- QUALCOMM, “Adaptive Design Solutions for VLSI Circuits,” \$50,000, gift, 09/01/09
- National Science Foundation (NSF), “Reclaiming Moore’s Law through Ultra Energy Efficient Computing,” \$2,778,507, with \$643,700 to Blaauw, PI: Prof. David Blaauw, 9/2009-08/2014

- National Science Foundation (NSF), “Probabilistic Wearout in Nanoscale,” \$300,000, with \$150,000 to Blaauw, PI: Dennis Sylvester, 8/2008-7/2011
- IBM Corporation/Defense Advanced Research Projects Agency (DARPA), “Strained Si/SiGe/Ge Heterojunction Tunneling Transistor (HETT) with Steep Subthreshold Slope for Extremely Low Power Electronics,” \$17,971,252, with \$600,000 to Blaauw, PI: Steve Koester, 1/2008-12/2009
- BAE Systems/United States Army, “Center for Objective Microelectronics and Biomimetic Adaptive Technology (COM-BAT),” \$8,962,200 with \$700,000 to Blaauw, PI: Kamal Sarabandi, 5/2008-5/2013
- Intel Corporation, “Adaptive Digital Design in the Nanometer Regime,” \$100,000, gift, 3/2008-3/2010
- Sun Microsystems, “Robust Low Voltage SRAM Design,” \$150,000, gift, 9/2007-9/2010
- Intel Corporation, “Circuit and Microarchitectural Methods for Subthreshold Design,” \$40,000, gift, 7/2007
- MARCO/DARPA - Gigascale Systems Research Center (GSRC), “Elastic: An Adaptive Self-Healing Architecture for Unpredictable Silicon,” \$600,000 to Blaauw, PI: David Blaauw, 9/2006 - 9/2009
- Semiconductor Research Corporation (SRC), “A Design Optimization Framework for Process Variation Tolerance,” \$390,000, with \$195,000 to Blaauw, PI: Dennis Sylvester, 9/2006 - 8/2009
- Intel Corporation, “Circuit and Microarchitectural Methods for Subthreshold Design” \$40,000, gift, 7/2006
- Semiconductor Research Corporation (SRC), “CAD Solutions for Parametric Yield Optimization,” \$321,000, with \$160,000 to Blaauw, PI: Dennis Sylvester, Co-PI: David Blaauw, University of Michigan, 9/2005-7/2008
- Intel Corporation, “Circuit and Microarchitectural Methods for Subthreshold Design” \$40,000, gift, 7/2005
- NSF Engineering Research Center (ERC) for Wireless Integrated Micro Systems (WIMS), “Subthreshold Processor Design,” \$60,000 to Blaauw, PI: Kenneth Wise, 5/2005-5/2010
- ARM, Ltd, “Low Power Computing for Embedded Applications,” \$5,000,000, with approx. \$1,600,000 to Blaauw, PI: Trevor Mudge, 5/2005-5/2010
- Semiconductor Research Corporation (SRC), “Optimization of Lithographic Induced Variability for Improved Circuit Performance,” \$161,029 to Blaauw, PI: David Blaauw, 9/2004-8/2007
- Intel Corporation, “Power Grid Integrity Analysis,” \$50,000, gift, 7/2004
- Photonics, Inc. \$75,000, gift, 6/2004-5/2005
- ARM, Ltd, “Low Power Computing for Embedded Applications,” \$240,000, with \$60,000 to Blaauw, PI: Trevor Mudge, 5/2004-5/2005
- National Science Foundation (NSF), Information Technology Research (ITR), “Collaborative Research ITR: Mobile Supercomputing,” \$1,900,000, with \$320,603 to Blaauw, PI: Trevor Mudge, 11/2003-11/2007
- Intel Corporation, “VLSI Design Curriculum,” \$247,292, with \$61,823 to Blaauw, PI: Richard Brown, 10/2003-10/2004
- MARCO/DARPA - Gigascale Systems Research Center (GSRC), “Power Aware Systems,” \$600,000, PI: David Blaauw, 9/2003-9/2006

- IBM Corporation, Center for Advanced Studies, “Static Performance Analysis under Process and Environment Variations,” \$40,000, Faculty Award, 9/2003
- Intel Corporation, “Power Grid Integrity Analysis,” \$50,000, gift, 7/2003
- Semiconductor Research Corporation (SRC), “Analysis and Reduction of Simultaneous Gate-Oxide Tunneling and Subthreshold Leakage Current,” \$360,000, with \$160,000 to Blaauw, PI: David Blaauw, 7/2003-7/2006
- National Science Foundation (NSF), “Performance Analysis and Optimization for Nanometer Design,” \$375,000, PI: David Blaauw, 6/2003-6/2006
- ARM, Ltd, “Low Power Computing for Embedded Applications,” \$240,000, with \$60,000 to Blaauw, PI: Trevor Mudge, 5/2003-5/2004
- IBM Corporation, Center for Advanced Studies, “Leakage Characterization and Analysis,” \$40,000, Faculty Award, 9/2002
- National Science Foundation (NSF), Information Technology Research (ITR), “Methodologies for Robust Design of Information Systems under Multiple Sources of Uncertainty”, \$1,800,00 with \$450,000 to Blaauw, PI: David Blaauw, 8/2002-8/2006
- Intel Corporation, “Power Grid Integrity Analysis,” \$50,000, gift, 7/2002
- MARCO/DARPA - Giga-Scale Research Center (GSRC), “Power Management for Nanometer design,” \$197,000, PI: David Blaauw, 10/2001-8/2003
- Semiconductor Research Corporation (SRC), “Variability in Chip-Level Performance and Signal Integrity Verification,” \$257,000, PI: David Blaauw, 10/2001-10/2004

IV Teaching Experience

<u>Semester</u>	<u>Class</u>	<u>Course Number</u>	<u>Size</u>	<u>Rating (out of 5) Course/Instructor</u>
Winter 2024	Advanced VLSI Design	EECS 627	40	3.3/3.9
Fall 2023	Advanced VLSI Design II	EECS 628	18	5/5
Winter 2023	Advanced VLSI Design	EECS 627	47	4.8/4/9
Fall 2022	VLSI Design I	EECS 427	53	4.9/4.9
Winter 2021	Advanced VLSI Design	EECS 627	27	4.3/4.2
Winter 2020	Advanced VLSI Design	EECS 627	26	4.5/4.3
Winter 2019	Advanced VLSI Design	EECS 627	24	4.5/4.6
Winter 2018	Advanced VLSI Design	EECS 627	18	4.08/4.75
Fall 2016	VLSI Design I	EECS 427	36	4.85/4.91
Fall 2015	Digital Integrated Circuits	EECS 312	35	4.58/4.81
Fall 2014	Advanced VLSI Design II	EECS 628	26	4.86/4.90
Winter 2014	Advanced VLSI Design	EECS 627	38	4.77/4.77
Fall 2013	VLSI Design I	EECS 427	39	4.74/4.78
Winter 2013	Introduction to Electronic Circuits	EECS 215	120	3.56/4.09
Winter 2012	Advanced VLSI Design	EECS 627	37	4.87/4.87
Fall 2011	VLSI Design I (Section 2)	EECS 427	18	4.79/4.79
Fall 2011	VLSI Design I (Section 1)	EECS 427	35	4.83/4.83
Winter 2011	Advanced VLSI Design	EECS 427	9	4.88/4.88
Fall 2010	Advanced VLSI Design II	EECS 628	19	4.81/5.00
Winter 2010	Advanced VLSI Design	EECS 627	19	4.85/4.96
Winter 2009	Advanced VLSI Design	EECS 627	23	4.75/4.75
Fall 2008	VLSI Design I	EECS 427	28	4.67/4.56

Winter 2007	Advanced VLSI Design	EECS 627	20	4.79
Fall 2006	VLSI Design I	EECS 427	31	4.89
Winter 2006	Advanced VLSI Design	EECS 627	22	4.55
Fall 2005	Topics in VLSI Design	EECS 598	12	4.25
Winter 2005	Advanced VLSI Design	EECS 627	20	4.79
Winter 2004	Advanced VLSI Design	EECS 627	35	4.59
Fall 2003	Introduction to Logic Design	EECS 270	87	4.77
Winter 2003	Advanced VLSI Design	EECS 627	36	4.61
Fall 2002	Introduction to Logic Design	EECS 270	109	4.77
Winter 2002	Advanced VLSI Design	EECS 627	40	4.31
Fall 2001	Issues in High-Performance Deep-Submicron Design	EECS 598	11	4.75

V Publications

A. Books

1. Ashish Srivastava, Dennis Sylvester and David Blaauw, Statistical Analysis and Optimization for VLSI: Timing and Power, Kluwer Academic Publishers, 2005

B. Book Chapters

1. Tutu Ajayi, Sumanth Kamineni, Morteza Fayazi, Yaswanth K. Cherivirala, Kyumin Kwon, Shourya Gupta, Wenbo Duan, Jeongsup Lee, Chien-Hen Chen, Mehdi Saligane, Dennis Sylvester, David Blaauw, Ronald Dreslinski Jr, Benton Calhoun, David D. Wentzloff, “Fully-Autonomous SoC Synthesis Using Customizable Cell-Based Analog and Mixed-Signal Circuits Generation,” in *IFIP/IEEE International Conference on Very Large Scale Integration - System on a Chip*, K. A. A. Makinwa, A. Baschiroto, and P. Harpe, Eds. Springer International Publishing, 2021
2. Sechang Oh, Yao Shi, Gyouho Kim, Yejoong Kim, Taewook Kang, Seok Hyeon Jeong, Dennis Sylvester, David Blaauw, “Low-Power Resistive Bridge Readout Circuit Integrated in Two Millimeter-Scale Pressure-Sensing Systems,” in *Low-Power Analog Techniques, Sensors for Mobile Devices, and Energy Efficient Amplifiers: Advances in Analog Circuit Design 2018*, K. A. A. Makinwa, A. Baschiroto, and P. Harpe, Eds. Springer International Publishing, 2019
3. Sechang Oh, Wanyeong Jung, Hyunsoo Ha, Jae-Yoon Sim, David Blaauw, “Energy-Efficient CDCs for Millimeter Sensor Nodes,” in *Efficient Sensor Interfaces, Advanced Amplifiers and Low Power RF Systems: Advances in Analog Circuit Design 2015*, K. A. A. Makinwa, A. Baschiroto, and P. Harpe, Eds. Springer International Publishing, 2016
4. Shidhartha Das, David Roberts, David Blaauw, David Bull, Trevor Mudge, “Architectural Techniques for Adaptive Computing”, Chapter in *Adaptive Techniques for Dynamic Processor Optimization: Theory and Practice*, Alice Wang and Sam Naffziger, editors, Springer Publishing Company, 2008
5. David Blaauw, Sanjay Pant, Rajat Chaudhry and Rajendran Panda, “Design and Analysis of Power Supply Networks,” Chapter in *Electronic Design Automation for Integrated Circuits Handbook*, Louise Sheffer, Luciano Lavagno and Grant Martin, editors, CRC Press, 2005
6. Sarvesh Kulkarni, Ashish Srivastava, Dennis Sylvester, David Blaauw, “Power Optimization Techniques using Multiple Supply Voltages,” Chapter in *Closing the Power Gap between ASIC and Custom*, David Chinnery and Kurt Keutzer, editors, Kluwer Academic Publishers, 2005
7. Dongwoo Lee, Bo Zhai, David Blaauw, Dennis Sylvester, “Static Leakage Reduction through Simultaneous V_t/T_{ox} and State Assignment,” Chapter in *Ultra Low-Power Electronics and Design*, Enrico Macii, editor, Kluwer Academic Publishers, 2004
8. David Blaauw, Abhijit Dharchoudhury, Rajendran Panda, “Design and Analysis of Power Distribution Networks for Processor Design,” Chapter in *IEEE Design of High Performance Microprocessors Circuits*, Anantha Chandrakasan, William Bowhill, and Frank Fox, editors, IEEE Press, 2000
9. Abhijit Dharchoudhury, Shantanu Ganguly, David Blaauw, “Timing and Signal Integrity Analysis,” Chapter in *Handbook for VLSI Design*, Wai Kai Chen, editor, IEEE Press, 2000

C. Invited Articles

1. Yufeng Gu, Arun Bala Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath N Paul, Md Vasimuddin, Sanchit Misra, David Theodore Blaauw, Satish Narayanasamy, Reetuparna

- Das, “GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis,” Communication of the ACM (CACM), accepted, Invited Paper, October 2024
2. Li Xu, David Blaauw, Dennis Sylvester, “Ultra-Low Power 32kHz Crystal Oscillators: Fundamentals and Design Techniques,” *Open Journal of the Solid-State Circuits Society*, Vol. 1, September 2021, pgs. 79-93
 3. Li Xu, Jeongsup Lee, Mehdi Saligane, David Blaauw, Dennis Sylvester, “Design Techniques of Integrated Power Management Circuits for Low Power Edge Devices,” IEEE Custom Integrated Circuits Conference (CICC), April 2021
 4. Sechang Oh, Minchang Cho, Xiao Wu, Yejoong Kim, Li-Xuan Chuo, Wootack Lim, Pat Pannuto, Suyoung Bang, Kaiyuan Yang, Hun-Seok Kim, Dennis Sylvester, David Blaauw, “IoT2 — the Internet of Tiny Things: Realizing mm-Scale Sensors through 3D Die Stacking,” ACM/IEEE Design Automation and Test in Europe Conference (DATE), March 2019, pgs. 686-691
 5. David Blaauw, “Unlocking New IoT Application Domains Through Ultra-Low Power mm-Scale Sensor Node Design,” Keynote Address at ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), July 2018
 6. Taekwang Jang, Gyouho Kim, Benjamin Kempke, Michael Henry, Nikolaos Chiotellis, Carl Pfeiffer, Dongkwun Kim, Yejoong Kim, Zhiyoong Foo, Hyeongseok Kim, Anthony Grbic, Dennis Sylvester, Hun-Seok Kim, David Wentzloff, David Blaauw, “Circuit and System Designs of Ultra-low Power Sensor Nodes with illustration in a miniaturized GNSS Logger for Position Tracking: Part I—Analog Circuit Techniques,” *IEEE Transactions on Circuits and Systems I (TCAS-I)*, Vol. 64, No. 9, September 2017, pgs. 2237-2249
 7. Taekwang Jang, Gyouho Kim, Benjamin Kempke, Michael Henry, Nikolaos Chiotellis, Carl Pfeiffer, Dongkwun Kim, Yejoong Kim, Zhiyoong Foo, Hyeongseok Kim, Anthony Grbic, Dennis Sylvester, Hun-Seok Kim, David Wentzloff, David Blaauw, “Circuit and System Designs of Ultra-low Power Sensor nodes with Illustration in a Miniaturized GNSS Logger for Position Tracking: Part II—Data Communication, Energy Harvesting, Power Management and Digital Circuits,” *IEEE Transactions on Circuits and Systems I (TCAS-I)*, Vol. 64, No. 9, September 2017, pgs. 2250-2262
 8. Wanyeong Jung, Dennis Sylvester, David Blaauw, “Low-Power Switched-Capacitor Converter Techniques for Small IoT Systems,” European Conference on Circuit Theory and Design (ECTD), September 2017
 9. Taekwang Jang, Myungjoon Choi, Yao Shi, Inhee Lee, Dennis Sylvester and David Blaauw, “Millimeter-Scale Computing Platform for Next Generation of Internet of Things,” IEEE International Conference on RFID (RFID), May 2016
 10. Taekwang Jang, Seokhyeon Jeong, Myungjoon Choi, Wanyeong Jung, Gyouho Kim, Yen-Po Chen, Yejoong Kim, Wootack Lim, Dennis Sylvester, David Blaauw, “Key Building Blocks and Integration Strategy of a Miniaturized Wireless Sensor Node,” IEEE European Solid-State Circuits Conference (ESSCIRC), September 2015
 11. Nathaniel Pinckney, David Blaauw, Dennis Sylvester, “Low Power Near-Threshold Design,” IEEE Solid-State Circuits Magazine, June 2015
 12. Inhee Lee, Yejoong Kim, Suyoung Bang, Gyouho Kim, Hyunsoo Ha, Yen-Po Chen, Dongsuk Jeon, Seokhyun Jeong, Wanyeong Jung, Mohammad Hassan Ghaed, Zhiyoong Foo, Yoonmyung Lee, Jae-Yoon Sim, Dennis Sylvester, and David Blaauw, “Circuit Techniques for Miniaturized Bio-medical Sensors,” IEEE Custom Integrated Circuits Conference (CICC), September 2014
 13. David Blaauw, Dennis Sylvester, Prabal Dutta, Yoonmyung Lee, Inhee Lee, Sechang Bang, Yejoong Kim, Gyouho Kim, Pat Pannuto, Ye-Shang Kuo, Dongmin Yoon, Wanyeong Jung,

- ZhiYoong Foo, Yen-Po Chen, Seok Hyeon Jeong, Myungjoon Choi, "IoT Design Space Challenges: Circuits and Systems" 2014 IEEE Symposium on VLSI Technology, June 2014
14. Yoonmyung Lee, Dennis Sylvester, David Blaauw, "Circuits for Ultra-Low Power Millimeter-Scale Sensor Nodes," 2012 Asilomar Conference on Signals, Systems and Computers (Asilomar), November 2012
 15. David Blaauw, Dennis Sylvester, Yoonmyung Lee, Inhee Lee, Suyoung Bang, Yejoong Kim, Gyouho Kim, Hassan Ghaed, "From Digital Processors to Analog Building Blocks: Enabling New Applications through Ultra-Low Voltage Design," **Invited paper** to the IEEE Subthreshold Microelectronics Conference (SubVt), Plenary Keynote, October 2012
 16. Nathaniel Pinckney, Korey Sewell, Ronald Dreslinski, Dave Fick, David Blaauw, Dennis Sylvester, Trevor Mudge, "Assessing the Performance of Parallelized Near-Threshold Computing," ACM/IEEE Design Automation Conference (DAC), June 2012
 17. Yoonmyung Lee, YeJoong Kim, Dongmin Yoon, David Blaauw, Dennis Sylvester, "Circuit and System Design Guidelines for Ultra-Low Power Sensor Nodes," ACM/IEEE Design Automation conference (DAC), June 2012
 18. Yoonmyung Lee, Dennis Sylvester, David Blaauw, "Synchronization of Ultra-Low Power Wireless Sensor Nodes", *IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, August 2011
 19. Gregory Chen, Scott Hanson, David Blaauw, Dennis Sylvester, "Circuit Design Advances for Wireless Sensing Applications," Proceedings of the IEEE, Special Issue on Wireless Sensor Networks, Vol. 98, No. 11, November 2010, pg. 1808 - 1827
 20. Prashant Singh, Dennis Sylvester, David Blaauw, "Adaptive Sensing and Design for Reliability," IEEE International Reliability Physics Symposium, May 2010
 21. Ronald G. Dreslinski, Michael Wieckowski, David Blaauw, Dennis Sylvester, Trevor Mudge, "Near-Threshold Computing: Reclaiming Moore's Law Through Energy Efficient Integrated Circuits," Proceedings of the IEEE, Special Issue on Ultra-Low Power Circuit Technology, Vol. 98, No. 2, February 2010, pg. 253 - 266
 22. Prashant Singh, Cheng Zhuo, Eric Karl, David Blaauw, Dennis Sylvester, "Sensor Driven Reliability and Wearout Management," *IEEE Design & Test of Computers (D&T)*, Vol. 26, No. 6, November/December 2009, pg. 40 - 49
 23. David Blaauw, Shidhartha Das, "CPU, Heal Thyself," IEEE Spectrum, August 2009
 24. Shidhartha Das, David Blaauw, David Bull, Krisztian Flautner, Rob Aitken, "Addressing Design Margins through Error-tolerant Circuits," ACM/IEEE Design Automation Conference (DAC), July 2009
 25. Shidhartha Das, David Blaauw, "Adaptive Design for Nanometer Technology," IEEE International Symposium on Circuits and Systems (ISCAS), May 2009
 26. Dennis Sylvester, Scott Hanson, Seok, Yu-Shiang Lin, David Blaauw, "Designing Robust Ultra-Low Power Circuits," International Electron Devices Meeting (IEDM), December 2008
 27. David Blaauw, Kaviraj Chopra, Ashish Srivastava, Lou Sheffer, "Statistical Timing Analysis: Basic Principles to State-of-the-Art," *Transactions on Computer-Aided Design of Integrated Circuits and Systems (T-CAD)*, invited review article, Vol. 27, No. 4, April 2008, pg. 589-607
 28. Scott Hanson, Bo Zhai, David Blaauw, Dennis Sylvester, "Energy-Optimal Circuit Design," IEEE International SoC Design Conference, November 2007

29. Sanjay Pant, Eli Chiprout, David Blaauw, "Power Grid Physics and Implications for CAD," *IEEE Design & Test of Computers (D & T)*, Vol. 24, No. 3, May-June 2007, pg. 246-254
30. Dennis Sylvester, Scott Hanson, Bo Zhai, and David Blaauw, "Design strategies for ultra-low voltage circuits," IEEE International SoC Design Conference, September 2006
31. Scott Hanson, Bo Zhai, David Blaauw, Dennis Sylvester, Andres Bryant, Xinlin Wang, "Energy Optimality and Variability in Subthreshold Design," ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED), September 2006
32. Shidhartha Das, David Roberts, Seokwoo Lee, Sanjay Pant, David Blaauw, Todd Austin, Trevor Mudge, Krisztián Flautner, "A Self-Tuning Dynamic Voltage Scaled Processor Using Delay-Error Detection and Correction," IEEE International Conference on Integrated Circuit Design & Technology (ICICDT), May 2006
33. David Blaauw and Bo Zhai, "Energy Efficient Design for Subthreshold Supply Voltage Operation," IEEE International Symposium on Circuits and Systems (ISCAS), May 2006
34. Rajeev R. Rao, David Blaauw, Dennis Sylvester, Anirudh Devgan, "Modeling and Analysis of Parametric Yield Under Power and Performance Constraints," *IEEE Design & Test of Computers (D&T)*, Vol. 22, No. 4, July-August 2005, pg. 376-385
35. Todd Austin, Valeria Bertacco, David Blaauw, Trevor Mudge, "Opportunities and Challenges for Better Than Worst-Case Design," ACM/IEEE Asia-Pacific Design Automation Conference (ASP-DAC), January 2005, pg. I-2
36. Bo Zhai, David Blaauw, Dennis Sylvester, Krisztián Flautner, "Extended Dynamic Voltage Scaling for Low Power Design," IEEE International SOC Conference, September 2004, pg. 389-394
37. Todd Austin, David Blaauw, Trevor Mudge, Krisztián Flautner, "Making Typical Silicon Matter with Razor" *IEEE Computer*, March 2004, pg. 57-65
38. David Blaauw, Kaushik Gala, "Inductance: Implications and Solutions for High-Speed Digital Circuits - Inductance Extraction and Modeling," IEEE International Solid-State Circuits Conference (ISSCC), February 2002, pg. 548-553
39. David Blaauw, "Signal Integrity Issues in High Performance Design," IEEE International Workshop on Power and Timing Modeling, Optimization and Simulation (Patmos), September 2001, pg. 5.1.1-5.1.4
40. Kaushik Gala, David Blaauw, Junfeng Wang, Vladimir Zolotov, Min Zhao, "Inductance 101: Analysis and Design Issues," ACM/IEEE Design Automation Conference (DAC), June 2001, pg. 329-334
41. David Blaauw, Kaushik Gala, Vladimir Zolotov, Rajendran Panda, Junfeng Wang, "On-Chip Inductance Modeling," ACM/IEEE Great Lake Symposium on VLSI Design (GLSVLSI), March 2000, pg. 75-80
42. David Blaauw, "Power Management Issues in High Performance Processor Design," IEEE Alessandro Volta Workshop on Low-Power Design (VOLTA), March 1999, pg. 2
43. David Blaauw, Abhijit Dharchoudhury, Rajendran Panda, Supamas Sirichotiyakul, Chanhee Oh, Tim Edwards, "Industrial Perspectives on Emerging CAD Tools for Low Power Processor Design," ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED), August 1998, pg. 143-148
44. Abhijit Dharchoudhury, Rajendran Panda, David Blaauw, Ravi Vaidyanathan, Bogdan Tutuianu, David Bearden, "Methodology for the Design and Analysis of Power Distribution Networks on the PowerPC Microprocessor," ACM/IEEE Design Automation Conference (DAC), June 1998, pg. 738-743

D. Journals

1. Kuan-Yu Chen, Thomas Mason Nelson, Alireza Khadem, Morteza Fayazi, Sanjay Sri Vallabh Singapuram, Ronald Dreslinski, Nishil Talati, Hun-Seok Kim, David Blaauw, “Canalis: A Throughput-Optimized Framework for Real-Time Stream Processing of Wireless Communication,” *ACM Transactions on Reconfigurable Technology and Systems*, accepted
2. Xiangdong Wei, Yufan Yue, Seungkyu Choi, Tutu Ajayi, Ronald Dreslinski, David Blaauw, Hun-Seok Kim, “A 33.06 Gb/s Reconfigurable Galois Field oFEC Decoder for Optical Inter-Satellite Communication,” *Journal: IEEE Solid-State Circuits Letters*, Vol. 7, October 2024, pgs. 331 – 334
3. Zichen Fan, Qirui Zhang, Hyochan An, Boxun Xu, Li Xu, Chien-wei Tseng, Yimai Peng, Andrea Bejarano-Carbo, Pierre Abillama, Ang Cao, Bowen Liu, Changwoo Lee, Zhehong Wang, Hun Seok Kim, David Blaauw, Dennis Sylvester, “AIMMI: Audio and Image Multi-Modal Intelligence via a Low Power SoC with 2M-Byte On-chip MRAM for IoT Devices,” *IEEE Journal of Solid-State Circuits (JSSC)*, Vol. 59, No. 10, pgs. 3488 – 3501, October 2022
4. Qirui Zhang, Zichen Fan, Hyochan An, Zhehong Wang, Ziyun Li, Guanru Wang, Pierre Abillama, Hun-Seok Kim, David Blaauw and Dennis Sylvester, “RoboVisio: A Micro-Robot Vision Domain-Specific SoC for Autonomous Navigation Enabling Fully-on-Chip Intelligence via 2MB eMRAM,” *IEEE Journal of Solid-State Circuits (JSSC)*, Vol. 59, No. 8, pgs. 2644 – 2658, August 2024
5. Kuan-Yu Chen, Chi-Sheng Yang, Yu-Hsiu Sun, Chien-Wei Tseng, Morteza Fayazi, Xin He, Siying Feng, Yufan Yue, Trevor Mudge, Ronald Dreslinski, Hun-Seok Kim, David Blaauw, “DAP: A 507 GMACs/J 256-Core Domain Adaptive Processor for Wireless Communication and Linear-Algebra Kernels in 12nm FINFET,” *IEEE Journal of Solid-State Circuits (JSSC)*, June 2022
6. Chien-Wei Tseng, Zhen Feng, Zichen Fan, Hyochan An, Yunfan Wang, Hun-Seok Kim, David Blaauw, “A Low-Power Highly Reconfigurable Analog FIR Filter With 11-bit Charge-domain DAC for Narrowband Receivers,” *IEEE Journal of Solid-State Circuits (JSSC)*, **Invited Paper** to the *Special Issue on VLSI 2023*, February 2024
7. Heejin Yang, Ji-Hwan Seol, Rohit Rothe, Zichen Fan, Qirui Zhang, Hun-Seok Kim, David Blaauw, Dennis Sylvester, “A 1.5 μ W Fully-Integrated Keyword Spotting SoC in 28nm CMOS with Skip-RNN and Fast-Settling Analog Frontend for Adaptive Frame Skipping,” *IEEE Journal of Solid-State Circuits (JSSC)*, Vol. 59, No. 1, Pgs. 29 – 39, January 2024
8. Yimai Peng, Seokhyeon Jeong, Kyojin Choo, Yejoong Kim, Li-Yu Chen, Rohit Rothe, Li Xu, Ilya Gurin, Omid Oliaei, Matthew J Thompson, Stephen Bart, Peter Hartwell, David Blaauw, Dennis Sylvester, “An Ultra-Low-Power Triaxial MEMS Accelerometer with High-Voltage Biasing and Electrostatic Mismatch Compensation,” *IEEE Journal of Solid-State Circuits (JSSC)*, January 2024, pgs. 1-17
9. Jungho Lee, Joseph G. Letner, Jongyup Lim, Gabriele Atzeni, Jiawei Liao, Abhilasha Kamboj, Bhavika Mani, Seokhyeon Jeong, Yejoong Kim, Yi Sun, Beomseo Koo, Julianna Richie, Elena della Valle, Paras R. Patel, Dennis Sylvester, Hun-Seok Kim, Taekwang Jang, Jamie Phillips, Cynthia A. Chestek, James Weiland, David Blaauw, “A Sub-mm³ Wireless Neural Stimulator IC for Visual Cortical Prosthesis with Optical Power Harvesting and 7.5 kbps Data Telemetry,” *IEEE Journal of Solid-State Circuits (JSSC)*, **Invited Paper** to the *Special Issue on VLSI 2023*, January 2024, pgs. 1-13
10. Anhang Li, Jeongsup Lee, Prashansa Mukim, Brian D. Hoskins, Pragya Shrestha, David Wentzloff, David Blaauw, Dennis Sylvester, Mehdi Saligane, “A fully integrated, automatically generated DC-DC converter maintaining >75% efficiency from 398 K down to 23 K across wide load ranges in 12 nm FinFET,” *IEEE Solid-State Circuits Letters*, Vol. 7, January 2024, pgs. 4-45

11. Qirui Zhang, Hyochan An, Andrea Bejarano-Carbo, Hun-Seok Kim, David Blaauw, Dennis Sylvester, "An Ultra-Low-Power H.264/AVC Intra-Frame Image Compression Accelerator for Intelligent Event-Driven IoT Imaging Systems," *IEEE Solid-State Circuit Letters*, Vol. 7, December 2023, pgs. 30-33
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E. Conference Papers

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2. Joseph G. Letner, Michael Barrow, Eunseong Moon, Paras R. Patel, Julianna M. Richie, Jordan L. W. Lam, Jungho Lee, Abhilasha Kamboj, Bhavika Mani, Yi Sun, Gabriele Atzeni, Beomseo Koo, Dawen Cai, Dennis Sylvester, James D. Weiland, Hun-Seok Kim, Taekwang Jang, Jamie Phillips, David Blaauw, Cynthia A. Chestek. "Efficient, rapid, and minimally invasive implantation of individual non-functional motes with penetrating subcellular-diameter carbon fiber electrodes," Society for Neuroscience Conference 2023, November, 2023
3. Pierre Abillama, Zichen Fan, Yu Chen, Hyochan An, Qirui Zhang, Seungkyu Choi, David Blaauw, Dennis Sylvester, Hun-Seok Kim, "SONA: An Accelerator for Transform-Domain Neural Networks with Sparse-Orthogonal Weights," 34th IEEE International Conference on Application-specific Systems, Architectures, and Processors (ASAP), July 2023, **Best Paper Award**
4. Zichen Fan, Qirui Zhang, Pierre Abillama, Sara Shoouri, Changwoo Lee, David Blaauw, Hun-Seok Kim, Dennis Sylvester, "TaskFusion: An Efficient Transfer Learning Architecture with Dual Delta Sparsity for Multi-Task Natural Language Processing," IEEE International Symposium on Computer Architecture (ISCA), June 2023
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28. Rohit Rothe, Minchang Cho, Kyojin Choo, Seokhyeon Jeong, Dennis Sylvester, David Blaauw, "A 192 nW 0.02 Hz High Pass Corner Acoustic Analog Front-End with Automatic Saturation Detection and Recovery," IEEE Symposium on VLSI Circuits (VLSI-Symp), **Invited Paper** to the *IEEE Journal of Solid-State Circuits (JSSC)*, *Special Issue on VLSI*, June 2021
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34. Kyojin Choo, Hyochan An, Dennis Sylvester, David Blaauw, “14.1-ENOB 184.9dB-FoM Capacitor-Array-Assisted Cascaded Charge-Injection SAR ADC,” IEEE International Solid-State Circuits Conference (ISSCC), February 2021
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 37. Jeongsup Lee, Yejoong Kim, Minchang Cho, Makoto Yasuda, Satoru Miyoshi, Masaru Kawaminami, David Blaauw, Dennis Sylvester, “A μ Processor Layer for mm-Scale Die-Stacked Sensing Platforms Featuring Ultra-Low Power Sleep Mode at 125°C,” IEEE Asian Solid-State Circuits Conference (A-SSCC), November 2020
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 49. Anuraag Soorishetty, Jian Zhou, Subhankar Pal, David Blaauw, Hun Seok Kim, Trevor Mudge, Ronald Dreslinski, Chaitali Chakrabarti, "Accelerating Linear Algebra Kernels on a Massively Parallel Reconfigurable Architecture," IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), May 2020
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 54. Li-Xuan Chuo, Yejoong Kim, Nikolaos Chiotellis, Makoto Yasuda, Satoru Miyoshi, Masaru Kawaminami, Anthony Grbic, David Wentzloff, Hun-Seok Kim, David Blaauw, "A 4 $\times$ 4 $\times$ 4-mm³ Fully Integrated Sensor-to-Sensor Radio using Carrier Frequency Interlocking IF Receiver with -94 dBm Sensitivity," IEEE Radio Frequency Integrated Circuits Symposium (RFIC), **Invited Paper** to the *IEEE Journal of Solid-State Circuits (JSSC), Special Issue on RFIC*, June 2019, **Best Paper Award**
 55. Najme Ebrahimi, Behzad Yektakhah, Kamal Sarabandi, Hun Seok Kim, David Wentzloff, David Blaauw, "A Novel Physical Layer Security Technique Using Master-Slave Full Duplex Communication," IEEE MTT-S International Microwave Symposium (IMS), June 2019

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222. Cheng Zhuo, Kanak Agarwal, Dennis Sylvester, David Blaauw, "Active Learning Framework for Post-Silicon Variation Extraction and Test Cost Reduction," ICCAD 2010, November 2010
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227. Mingoo Seok, David Blaauw, Dennis Sylvester, "Clock Network Design for Ultra-Low Power Applications," ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED), August 2010
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243. Ravikishore Gandikota, David Blaauw, Dennis Sylvester, "Interconnect Performance Corners considering Crosstalk Noise," IEEE International Conference on Computer Design (ICCD), October 2009
244. Yu-Shiang Lin, Dennis Sylvester, David Blaauw, "Near-Field Communications using Phase-Locking and Pulse Signalling for Millimeter-Scale Systems," IEEE Custom Integrated Circuits Conference (CICC), September 2009
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247. Ronald G. Dreslinski, David Fick, David Blaauw, Dennis Sylvester, Trevor Mudge, "Reconfigurable Multicore Server Processors for Low Power Operation," International Symposium on Systems, Architectures, Modeling and Simulation (SAMOS), July 2009
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73. "Logic Gate Size Optimization Process for an Integrated Circuit Whereby Circuit Speed is Improved While Circuit Areas is Optimized," Patent Number 5,619,418, issued on April 8, 1997
74. "Message Sequence Number Control in a Virtual Time System," Patent Number 5,617,561, issued on April 1, 1997

VI Scholarly Addresses

A. Conference Keynote Addresses and Invited Presentations

1. Invited presentation, “The Internet of Tiny Things - IoT²: Challenges and Opportunities in mm-Scale Computing,” ECE Distinguished Lecture Series, George Washington University, December 2018
2. Plenary Keynote Address, “Unlocking New IoT Application Domains Through Ultra-Low Power mm-Scale Sensor Node Design,” ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), July 2018
3. Invited presentation, “Low-Power Circuit Techniques for IoT Energy Harvesting,” ACM/IEEE International Symposium on Quality Electronic Design (ISQED), March 2016
4. Plenary Keynote Address, “From Digital Processors to Analog Building Blocks: Enabling New Applications through Ultra-Low Voltage Design,” IEEE Subthreshold Microelectronics Conference (SubVt), October 2012
5. Invited presentation, “Adaptive Sensing and Design for Reliability,” IEEE International Reliability Physics Symposium, May 2010
6. Invited presentation, “Architectural Techniques for Self-Adaptive Computing,” IEEE International Solid-State Circuits Conference (ISSCC), February 2007
7. Invited presentation, “Energy Optimality and Variability in Subthreshold Design,” ACM/IEEE International Symposium on Low-Power Electronics and Design (ISLPED), September 2006
8. Invited presentation, “Energy Efficient Design for Subthreshold Supply Voltage Operation,” IEEE International Symposium on Circuits and Systems (ISCAS), May 2006
9. Invited presentation, “Extended Dynamic Voltage Scaling for Low Power Design,” IEEE International SOC Conference, September 2004
10. Invited presentation, “Signal Integrity Issues in High Performance Design,” IEEE International Workshop-Power and Timing Modeling, Optimization and Simulation (Patmos), Switzerland, September 2001
11. Invited presentation, “Inductance 101: Analysis and Design,” ACM/IEEE Design Automation Conference, June 2001
12. Invited presentation, “Inductance Extraction and Modeling,” ACM/IEEE Great Lakes Symposium on VLSI Design (GLSVLSI), March 2000
13. Keynote address, “Power Management Issues in High Performance Processor Design,” IEEE Alessandro Volta Workshop on Low-Power Design (VOLTA), Italy, March 1999
14. Keynote address, “Industrial Perspectives on Emerging CAD Tools for Low Power Processor Design,” ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), August 1998

VII Professional Activities

A. Professional Societies

- Fellow of the Institute of Electrical and Electronics Engineers (IEEE).
- Member of the Association of Computing Machinery (ACM).

B. Editor, Co-Editor, and Associate Editor Positions

- Associate editor, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, December 2003 - January 2006
- Co-guest editor, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, special issue on the Design Automation Conference, 2002
- Co-guest editor, *IEEE Design & Test of Computers*, special issue on the Design Automation Conference, 2002
- Co-guest editor, *IEEE Transactions on Very Large Scale Integration Systems (T-VLSI)*, special issue on Low Power Electronics, 1999

C. Conference and Workshop Organization

- Member, technical program committee, IEEE International Solid-State Circuits Conference (ISSCC), 2018 - current
- Member, technical program committee, IEEE International Solid-State Circuits Conference (ISSCC), 2006 – 2009
- Member, technical program committee, ACM/IEEE Workshop on Timing in Synthesis and Specification (TAU), 2004 - 2007
- Member, executive committee, ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), 1999 - 2006
- Member, technical program committee, ACM/IEEE Design Automation Conference (DAC), 1997, 2005 – 2006
- Member, technical program committee, ACM/IEEE International Conference on Computer-Aided Design (ICCAD), 2002 - 2004
- Member, executive committee, ACM/IEEE Design Automation Conference (DAC), 2001 - 2003
- Panel Chair, ACM/IEEE Design Automation Conference (DAC), 2003
- Co-Chair, technical program committee, ACM/IEEE Design Automation Conference (DAC), 2001 - 2002
- General Co-Chair, ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), 2000
- Tutorial Chair, ACM/IEEE Design Automation Conference (DAC), 2000
- Co-Chair, technical program committee, ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED), 1999

D. Consulting & Advisory Boards

- **Technical Consulting and Advisory boards**
 - Member of University of Illinois at Urbana – Champaign Advisory Panel 2013 – current

- Gear Inc. 2013 – 2015
- Apache Design Automation – member of advisory board
- Nascentric, Technical Consulting, 2008
- CLK Design Automation (CLK-DA), Technical Consulting, 2005 – 2008
- **Legal Consulting**
 - Parkins Coie LLP, 2014 – 2015
 - WilmerHale, 2012 – 2013
 - Alston & Bird, 2010 – 2011
 - Weil, Gotshal & Manges, 2008 – 2010
 - WilmerHale, 2007
- F. Refereeing and Reviewing**
 - NSF, SRC, Natural Science and Engineering Research Council of Canada (NSERC)
 - IEEE, IEEE T-VLSI, ACM TODAES, IEEE D&T
 - DAC, ICCAD, ISLPED, ICCD, ISPD, TAU, DATE, ISCAS, ISQED, PACS